

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY - UI DESIGN Lab Internal -1 - B.Tech V Sem IT A.Y: 2024-25

1.
 - a. Install Flutter and Dart SDK **CO1,L5**
 - b. Explore various Flutter widgets (Text, Image, Container, etc.) **CO1,L5**
 - c. What is UI design **CO1,L5**
2.
 - a. Write a simple Dart program to understand the language basics **CO2,L5**
 - b. Implement different layout structures using Row, Column, and Stack widgets **CO2,L5**
 - c. What is flutter
3.
 - a. Learn about stateful and stateless widgets. **CO3,L5**
 - b. What is dart why it is used in flutter **CO3,L5**
4.
 - a. Install Flutter and Dart SDK **CO1,L5**
 - b. Explore various Flutter widgets (Text, Image, Container, etc.) **CO1,L5**
 - c. What are elements of flutter **CO1,L5**
5.
 - a. Write a simple Dart program to understand the language basics **CO2,L5**
 - b. Implement different layout structures using Row, Column, and Stack widgets **CO2,L5**
 - c. What is widget explain with example
6.
 - a. Learn about stateful and stateless widgets. **CO3,L5**
 - b. What is state full widget explain with example **CO3,L5**
7.
 - a. Write a simple Dart program to understand the language basics **CO2,L5**
 - b. Explore various Flutter widgets (Text, Image, Container, etc.) **CO2,L5**
 - c. What is stateless widget explain with example **CO2,L5**
8.
 - a. Learn about stateful widgets. **CO2,L5**
 - b. Implement different layout structures using Row, Column, and Stack widgets **CO2,L5**
 - c. What is UI design
9.
 - a. Learn about stateless widgets. **CO1,L5**
 - b. Write a simple Dart program to understand the language basics **CO1,L5**
 - c. What is flutter **CO1,L5**
10.
 - a. Explore various Flutter widgets (Text, Image, Container, etc.) **CO2,L5**
 - b. Learn about stateful widgets **CO2,L5**

FLUTTER LAB INTERNAL-2

1. a) Design a responsive UI that adapts to different screen sizes. CO3, L5
b) Implement media queries and breakpoints for responsiveness CO3, L5
c) **What is the difference between MaterialApp and WidgetsApp in Flutter?**
2. a) Set up navigation between different screens using Navigator. CO4, L5
b) Implement navigation with named routes. CO4, L5
c) **What is the difference between Navigator and Router in Flutter?**
3. a) Implement state management using set State and Pr|ovider. CO4, L5
b) Experiment with different types of animations (fade, slide) CO5, L5
c) **What is the difference between push and pushReplacement methods in Flutter?**
4. a) Create custom widgets for specific UI elements. CO3, L5
b) Apply styling using themes and custom styles. CO3, L5
c) **What is the purpose of the Widget Inspector in Flutter?**
5. a) Design a form with various input fields. CO4, L5
b) Implement form validation and error handling. CO4, L5
c) **What is the difference between GridView and ListView in Flutter?**
6. a) Add animations to UI elements using Flutter's animation framework. CO5, L5
b) Experiment with different types of animations (fade, slide) CO5, L5
c) **What is the purpose of the InheritedWidget widget in Flutter?**

Mahatma Gandhi Institute of Technology

Department of Information Technology

Class : B.Tech Sem : V

Branch : IT

Date(s) :

18.12.2024

Academic Year : 2024 – 2025

Max. Marks : 60

M

Course : UI design- Flutter Lab

Course Code :

CS553PC

EXTERNAL LAB

1. a) Install Flutter and Dart SDK.

1 b) Write a simple Dart program to understand the language basics.

CO1,L5

2. a) Explore various Flutter widgets (Text, Image, Container, etc.).

b) Implement different layout structures using Row, Column, and Stack widgets.

CO2,L5

3. a) Design a responsive UI that adapts to different screen sizes.

b) Implement media queries and breakpoints for responsiveness.

CO1,L5

4. a) Set up navigation between different screens using Navigator.

b) Navigation with named routes.

CO2,L5

5. a) Learn about state full and stateless widgets.

b) Implement state management using set State and Provider

CO2,L5

6. a) Create custom widgets for specific UI elements.

b) Apply styling using themes and custom styles.

CO3,L5

7. a) Design a form with various input fields.

b) Implement form validation and error handling.

CO4,L5

8. a) Add animations to UI elements using Flutter's animation framework.

b) Experiment with different types of animations (fade, slide, etc.).

CO5,L5